

FIRST-MOVE ADVANTAGE IN ONLINE CHESS ACROSS RATING BRACKETS

Does White's edge from moving first change with player skill and time control? Answered with all 86.5 million rated Lichess games from June 2026.

INTRODUCTION AND MOTIVATION

Chess players have known for over a century that White enjoys a first-move advantage, but does that edge hold at every skill level, or is it something only strong players take advantage of? And does it change with time control? Excel and pandas both break well before you can answer that question at scale, so we profiled, broke, and then scaled past the full Lichess rated games dump for June 2026 with command-line tools and PySpark on Google Cloud Dataproc.

PROFILING

Dataset Overview:

- Size: ~148 GB uncompressed
- Rows: 1.7 billion (from wc -l count)

Characterized the raw dataset with

CLI tools only: du, wc -l, head/tail, cut, sort | uniq -c, awk, grep -c — every command streamed through zstdcat and timed with time

Key Data Insights:

- Average White Elo: 1643.37 computed in 14m 24s
- Top opening: Queen's Pawn Game (2.29M games)
- Cheapest command (grep -c, 14m24s) to most expensive (full-file awk aggregate, 1h22m (single core))

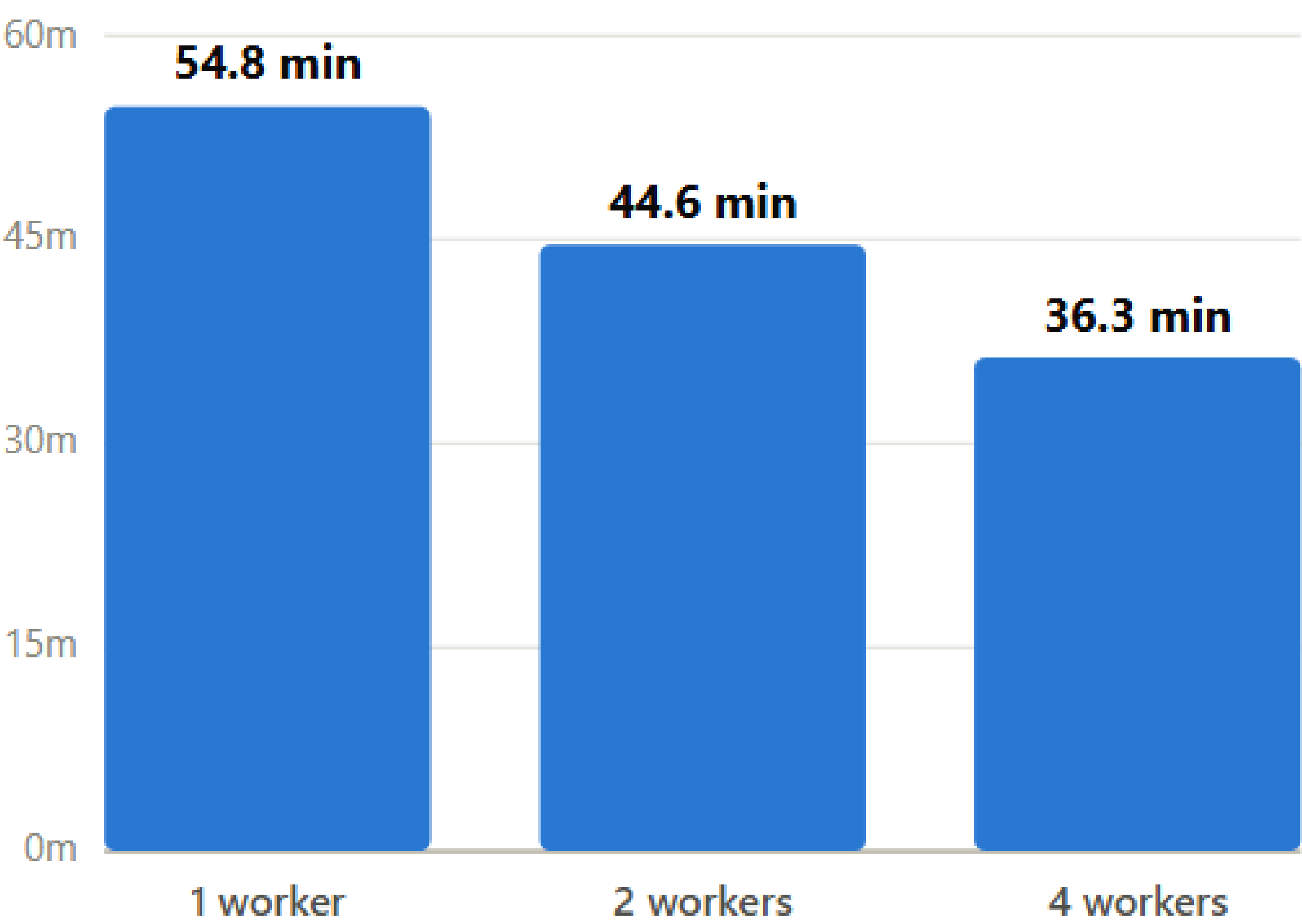
We **never** used RAM here.

Cost of Sequential Streaming:

- Reading the start: 0.03 seconds
- Reading the end: 59.7 minutes
- Sorting openings: 22.2 minutes

SCALING

The Same PySpark Job (explicit schema, transforms, groupBy, a window function, a join function, caching, and a Spark SQL cross-check) reads the full dataset directly from GCS and runs unchanged on Dataproc clusters of 1, 2, and 4 workers



runtime only falls ~18.6% per doubling, not ~50% linear scaling would predict

BREAKING

Pandas Test

read_csv() with no chunk size on the ~8.9 GB decompressed PGN pulled the entire file into RAM before doing anything else. Peak RSS 8.80 GiB, then crashed on a bad byte after 346s

```
PS Z:\lichess_db_standard_rated_2026-06.pgn> python .\pandas_pgn_test.py ".\lichess_db_standard_rated_2026-06.pgn"
eout: 600
Pandas breaking-point experiment
Input: Z:\lichess_db_standard_rated_2026-06.pgn\lichess_db_standard_rated_2026-06.pgn
Raw file size: 8.913 GiB
Time limit: 600 seconds
Pandas process ID: 3216

Open Task Manager now and watch the Python process.

elapsed=0.1s pandas_rss=0.019 GiB system_used=19.494 GiB system_available=12.324 GiB memory_percent=61.3%
```

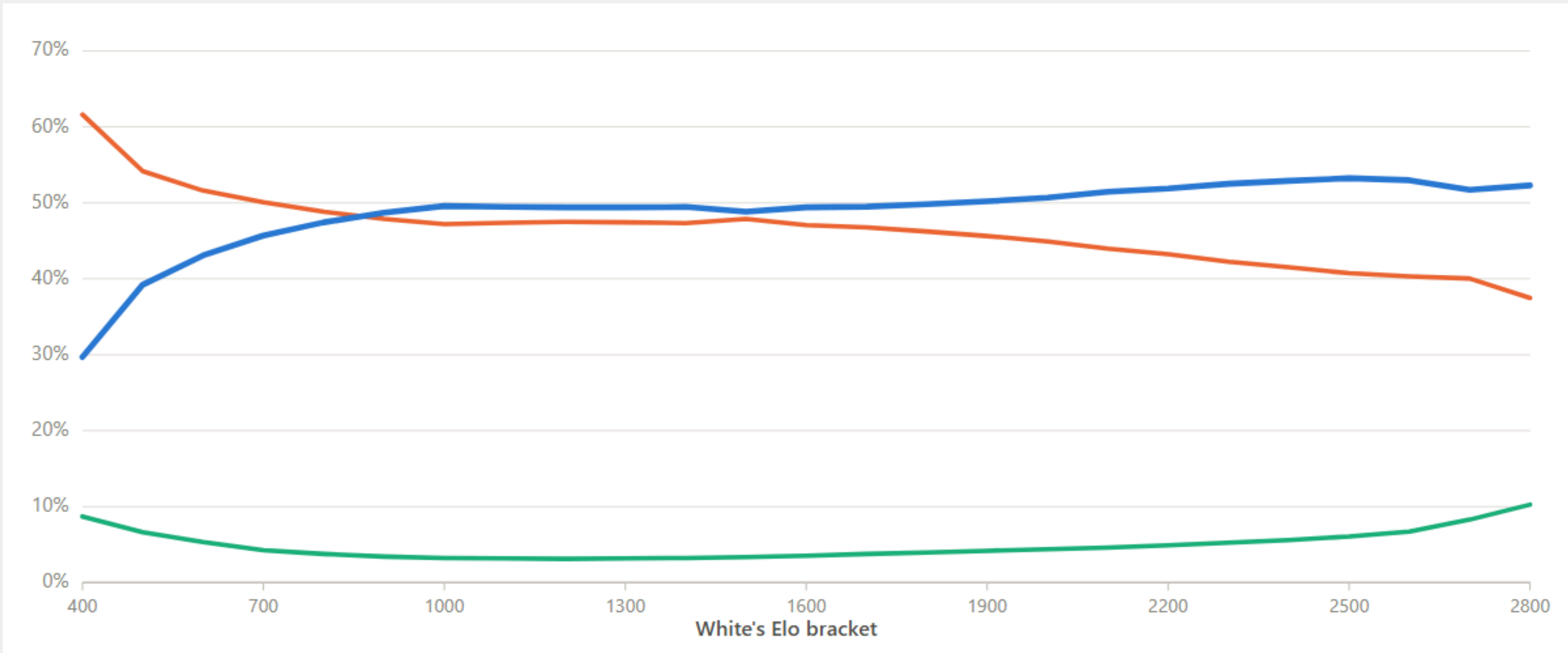
Task Manager: Python process at 8,980.8 MB, "Very high" power usage

Processes							
Name	Status	13% CPU	20% GPU	74% Memory	16% Disk	0% Network	Power usage
Python (2)		4.9%	0%	8,980.8 MB	72.2 MB/s	0 Mbps	Very high
Python		4.9%	0%	8,930.4 MB	72.2 MB/s	0 Mbps	Very high
Python		0%	0%	50.4 MB	0.1 MB/s	0 Mbps	Very low

CLI vs. Spark cross-check: grep -c counted 86,483,328 games in 14m24s; the Spark job independently landed on 86,458,368. This is a 0.03% gap explained by Spark's stricter row filter.

RESULTS/FINDINGS

White win % Black win % Draw %



Skill matters. White's overall edge barely moves but climbs steadily with player strength. Weak players convert the first-move advantage only weakly. At the lower brackets, Black actually wins more, likely from White blundering the extra tempo away.

AUTHORS

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GITHUB

<https://github.com/TrtQu/First-Move-Advantage-in-Online-Chess.git>

26 GB

compressed size (pgn.zst)

1.73B

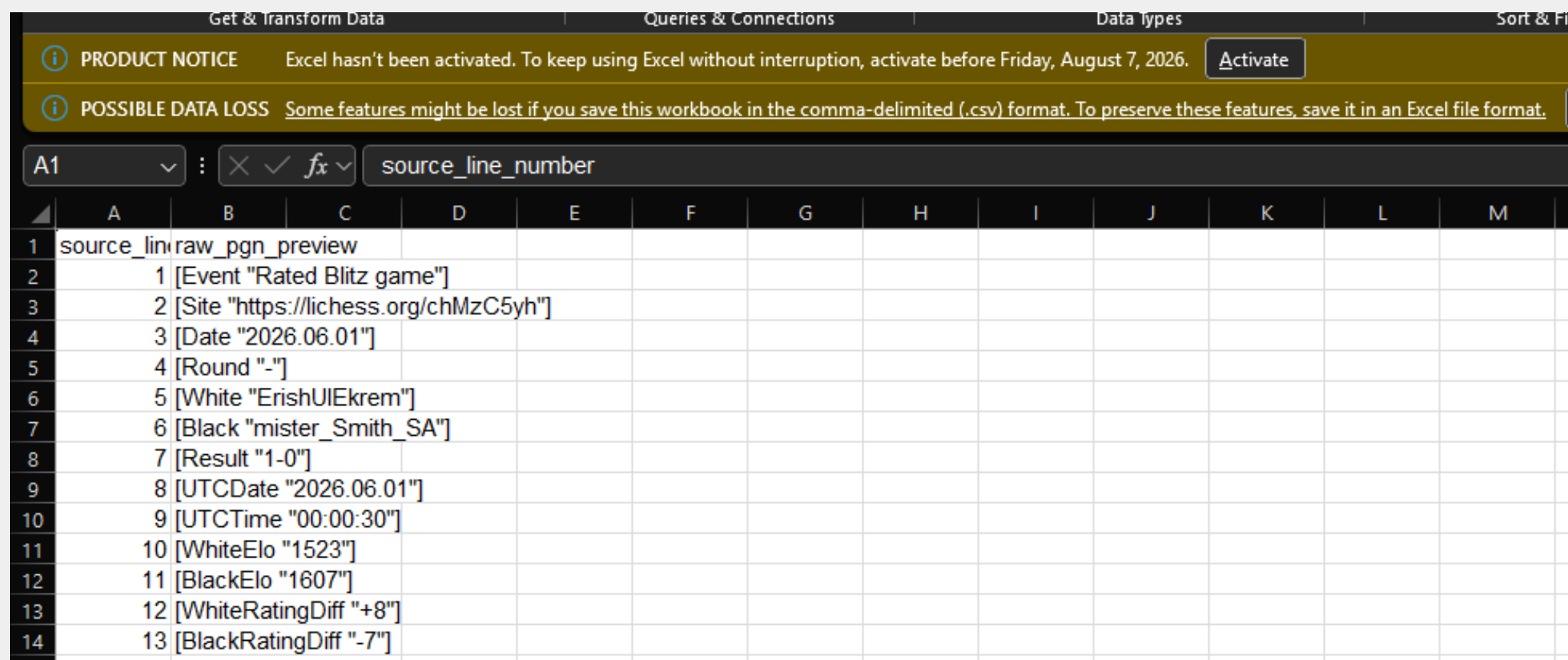
lines of data after decompression

86.48M

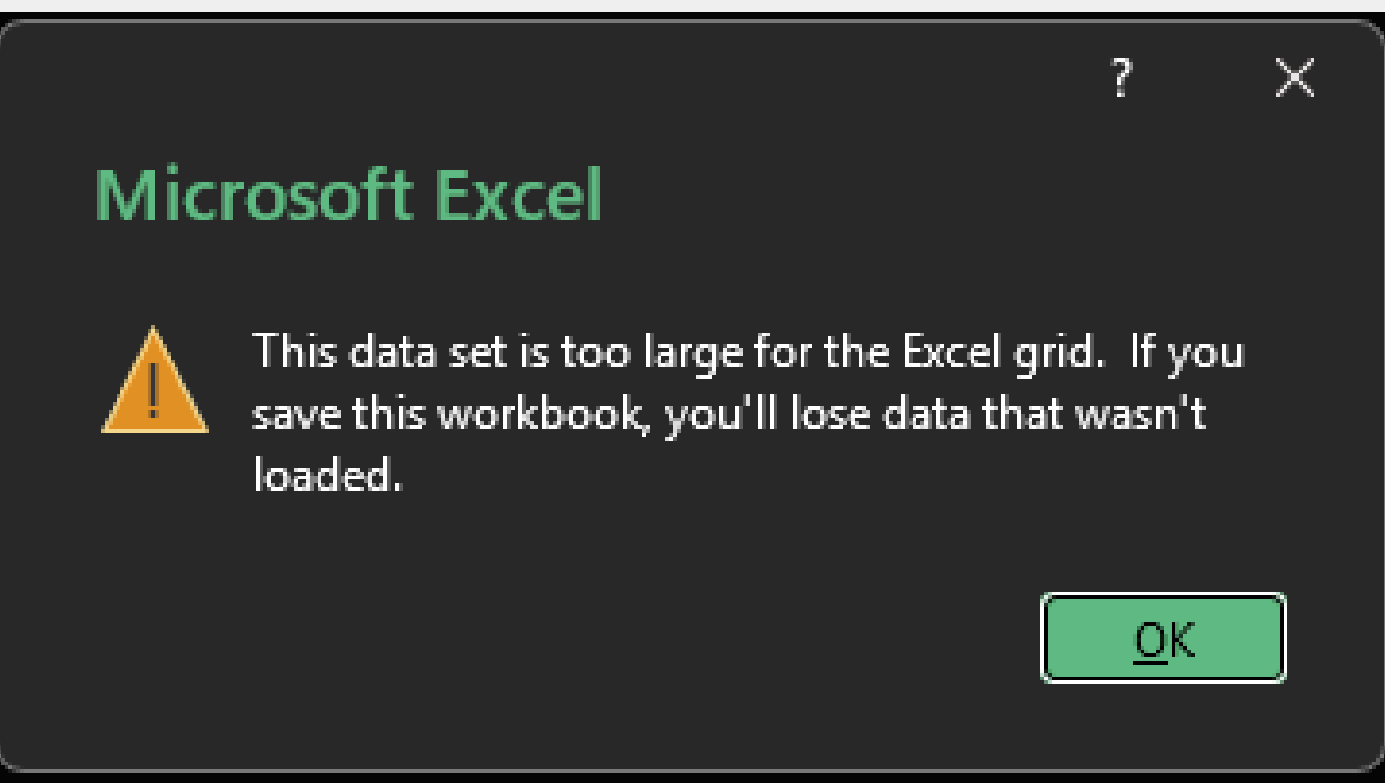
full rated chess games analyzed

Excel Test

A 400 MB/ 10,050,001 row CSV silently truncates to Excel's 1,048,576 row cap, then throws a warning.



exactly 1,048,576 rows loaded



"This data set is too large for the Excel grid"

29.7% → 52.9%

White's win rate at the lowest Elo bracket(400 vs. master level (2400+))

61.6% → 41.5%

Black's win rate falls by the mirror amount, crossing White's line at around 900-1000 Elo

Time control	White win %	Draw %	Games
Bullet	49.99%	2.87%	31.7M
Blitz	49.59%	4.32%	40.1M
Rapid	49.54%	4.66%	14.2M
Classical	48.82%	5.99%	383K

The best single opening for White by win rate:

Zukertort Opening: Tennison Gambit, Brigg's Trap,
76.0% white wins across 2,231 Blitz games.

The biggest factor that changes white's winning chances over Black's is skill - stronger players capitalize on the first move advantage far more.